



College of Computing and Information Sciences Department of Information Technology

Mini Data Analysis Project

Overview

In this project, students will apply concepts in descriptive statistics to analyze a real-world dataset. The goal is to transform raw data into meaningful insights using statistical tools and clear interpretation.

Students will **download or source a dataset**, formulate **5 research questions**, and answer them using appropriate descriptive statistical techniques.

Learning Objectives

By the end of this project, students should be able to:

- Apply measures of central tendency and variability
- Organize data using tables and graphs
- Interpret statistical results in context
- Formulate and answer data-driven questions
- Communicate findings effectively

Project Requirements

1. Dataset Selection

Students must:

- Download a dataset from sources such as:
 - Kaggle
 - Government open data portals
 - WHO / World Bank
- Dataset must have:
 - At least **30 observations**
 - At least **5 variables (columns)**

2. Research Questions (Required: 5 Questions)

Students must create **5 guide questions** that can be answered using descriptive statistics.

3. Data Processing

Students should:

- Clean the dataset (remove missing or inconsistent data)
- Encode categorical variables if needed
- Present a brief description of the dataset

4. Statistical Analysis

Students must compute and present:

Measures of Central Tendency

- Mean
- Median
- Mode

Measures of Variability

- Range
- Variance
- Standard Deviation

Data Distribution Tools

- Frequency Distribution Table
- Graphs (at least 2):
 - Bar Chart
 - Histogram
 - Pie Chart
 - Line Graph etc.

5. Interpretation of Results

For each of the 5 questions, students must:

- Present the computed result
- Provide a **clear interpretation in paragraph form**

Project Output Format

Students must submit:

Written Report (PDF or Word)

Sections:

1. Introduction
2. Dataset Description
3. Research Questions
4. Methodology
5. Results and Analysis

Data File

- Excel (.xlsx) or CSV file with computations

Suggested Tools

- Microsoft Excel
- Google Sheets
- SPSS (optional)

Grading Rubric (100 Points)

Criteria	Description	Points
Dataset Appropriateness	Relevance and completeness of data	15
Research Questions	Clarity and alignment with analysis	15
Statistical Analysis	Correctness of computations	25
Graphs & Tables	Clarity and accuracy of visuals	15
Interpretation	Depth and correctness of insights	20
Organization & Presentation	Structure, clarity, and neatness	10